

# Artifact-3

## Vision Document

### Revision History

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# **Vision 1.0**

## **1. Introduction**

Even more than any service ever would, the fast-food culture in Pakistan has now exceeded the necessary and become a necessity for busy families and working-class people in a fast-paced world. At this stage, it is fast food outlets with very generic profiles that are hardly customized, void of all cultural distinctions, and unconcerned about how the customers experience fast food outlets.

This is where **SpiceSync: Fast Food Management System** comes in. SpiceSync connects AI, a local ingredient and real-time services to create engaging fast food ordering experiences that are personalized, culturally grounded, and environmentally aware. SpiceSync will house the end-users in mobile applications and web experience to offer tailored meal recommendations, transparent ingredient sourcing, voice-assisted ordering, and other services.

### **1.1 Purpose**

The concept of **SpiceSync: Fast Food Management System** is to be an active and helpful way to modify the fast-food ordering experience for Pakistanis. Whereas, the existing apps are an example of the non-personal, culturally irrelevant, as well as non-sustainable aspects of food service applications. SpiceSync would aid customers with AI-driven meal recommendations for personalized regional flavors, real-time tracking, as well as green delivery for home and office, and embrace a local diversity of cooking qualities in addressing the digital convenience issue.

### **1.2 Scope**

The application will be web and mobile-based app for users from the urban areas of Pakistan. The system features social login user registration and personalize meal combo by AI; real-time order tracking; regional spice level suggestion; multilingual support - both Urdu and English, and also many more like display halal certification, payment options (JazzCash, Easypaisa, cash), special offers during festivals, loyalty rewards, voice-assisted ordering through the SpiceBot, etc. All these users will also benefit from the central operations of this platform, which is primarily used for direct consumers, partner restaurants, and delivery agents, thus being the center of fastfood management.

#### **Major Features:**

- User registration/login with social media integration (Google, Facebook).
- Meal recommendations based on preferences (Spicy, vegetarian, etc.).
- Real-time order tracking with delivery.
- Regional spice customization (e.g., Peshawari, Punjabi, Islamabad mild).
- Subscription plans for weekly office lunches.
- Multiple payment options (JazzCash, Easypaisa, cash-on-delivery).
- Loyalty reward points (spice points) for discounts.
- Voice-assisted ordering via “SpiceBot” (Urdu/English).
- Integration with Foodpanda/Cheetah for expanded delivery.

### 1.3 Definitions, Acronyms, and Abbreviations

- AI – Artificial Intelligence.
- UI – User Interface.

### 1.4 References

- **User Account and Data Management Requirements:**

This reference focuses on the requirements for user registration/login, user profile management, data storage, and data security.
- **Order Processing and Fulfillment Requirements:**

This reference outlines the requirements for order placement, order tracking, order status updates, and integration with delivery services.
- **Menu and Product Catalog Requirements:**

This reference details the requirements for managing the menu, including adding/editing items, categorizing food, handling variations, and displaying nutritional information.
- **Payment and Transaction Requirements:**

This reference specifies the requirements for processing payments, integrating multiple payment gateways, handling transactions securely, and managing loyalty programs.
- **User Interface and Experience Requirements:**

This reference defines the requirements for the user interface design, usability, accessibility, multilingual support, user feedback mechanisms, and promotional features.

## 1.5 Overview

In Pakistan's rapidly urbanizing society, fast food responds to very many, but the existing platforms fail to deliver personalized and localized experiences. SpiceSync bridges the divide with culturally authentic customized online ordering. This system will involve AI-based meal suggestions about user preferences, together with regional food flavoring. It is planned to be a flexible and user-oriented platform for families and office people in search of convenient and relevant fast-food options.

## 2. Positioning

### 2.1 Problem Statement

|                                |                                                                                                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The problem of                 | Online Food Management System                                                                                                                                                                                                                           |
| affects                        | Users, Restaurant Partner, Delivery Partner                                                                                                                                                                                                             |
| the impact of which is         | Lack cultural relevance, convenience, and personalization, leading to lower satisfaction and limited brand loyalty.                                                                                                                                     |
| a successful solution would be | A smart, culture-sensitive fast food management system that is powered by artificial intelligence to enhance customer satisfaction by appreciating the local food culture, providing real-time services and convenience to families and office workers. |

### 2.2 Product Position Statement

|                    |                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| For                | Users                                                                                                                                                        |
| Who                | Online Management System                                                                                                                                     |
| The (product name) | Is SpiceSync: Fast Food Management System.                                                                                                                   |
| That               | Provides AI-curated meal recommendations, real-time order tracking, regional spice customization, multilingual support, and flexible group ordering options. |
| Unlike             | Traditional food apps that offer generic menus, limited regional flavors, no meal personalization, and no proper subscription or group ordering options.     |
| Our product        | SpiceSync offers tailored food experiences based on user preferences, supports halal certification transparency, eco-                                        |

friendly deliveries, real-time GPS tracking, voice-assisted ordering, loyalty rewards (Spice Points), and festive promotions—available 24/7 via mobile and web platforms.

### 3. Stakeholder and User Descriptions

The SpiceSync Fast Food Management System is developed to provide personalized and culturally relevant food ordering services to users with modern features. The SpiceSync system has the following features that are given below.

- Provide login or signup facilities to user with social media integration.
- The system provides AI-curated meal combos based on user preferences.
- The system allows real-time order tracking with delivery.
- Users can customize regional flavors (e.g., Karachi spicy, Lahore creamy).
- The system provides subscription plans for weekly office lunches.
- Halal certification and ingredient transparency is maintained.
- Multiple payment options are available (JazzCash, Easypaisa, cash-on-delivery).
- Users can participate in recipe contests for menu inclusion.
- Loyalty reward points system ("spice points") offers discounts.
- Multilingual support in Urdu/English is provided.
- Nutritional information and calorie counters are available.
- Instant feedback rating system for orders is implemented.
- Group ordering with bulk discounts is possible.
- Festive promotions (Eid, Ramadhan etc.) are offered.
- Integration with Foodpanda/Cheetah for delivery.
- Voice-assisted ordering via "SpiceBot" is available.

#### 3.1 Stakeholder Summary

| Name                   | Description                                                 | Responsibilities                                                                                                                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Platform Owners</b> | This system evolved around the need of the platform owners. | <ul style="list-style-type: none"> <li>➤ Managing the services of users and partners.</li> <li>➤ Platform owners have an <i>ability</i> to update the system.</li> <li>➤ Checking the performance of restaurants and delivery personnel.</li> <li>➤ Responsible for maintaining AI-curated meal combos and SpiceBot.</li> </ul> |
| <b>Restaurant</b>      | They work under the                                         | <ul style="list-style-type: none"> <li>➤ Managing and updating food menus.</li> </ul>                                                                                                                                                                                                                                           |

|                           |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Partners</b>           | platform owner instructions.                                        | <ul style="list-style-type: none"> <li>➤ Maintaining the <i>halal certification</i> and transparency.</li> <li>➤ Participating in recipe contests for menu innovation.</li> <li>➤ Handling group orders and festive promotions.</li> </ul>                                                                                                                                                 |
| <b>Customers</b>          | The customers use the system to register or unregister themselves.  | <ul style="list-style-type: none"> <li>➤ After registration, customers have <i>responsibilities</i> to place orders and pay timely.</li> <li>➤ Customers have <i>responsibilities</i> to provide feedback and ratings.</li> <li>➤ Customers must maintain their account security (password, payment).</li> <li>➤ Customers can use SpiceBot for <i>voice-assisted</i> ordering.</li> </ul> |
| <b>Delivery Personnel</b> | The delivery personnel work under the order system of the platform. | <ul style="list-style-type: none"> <li>➤ Picking up and delivering orders safely.</li> <li>➤ Delivery personnel have a <i>responsibility</i> to update the order status.</li> <li>➤ Must handle group or festive bulk orders properly.</li> <li>➤ Coordinate with restaurant partners and platform owners when needed.</li> </ul>                                                          |

### 3.2 User Summary

| Name            | Description                                                                                                            | Responsibilities                                                                                                                                                                                               | Stakeholders |
|-----------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>User</b>     | Occasional users who order fast food for convenience and quick meals.                                                  | <ul style="list-style-type: none"> <li>➤ Browse menu, place orders, track delivery, give feedback.</li> <li>➤ Check nutritional info, review ingredient sourcing, rate healthy meals.</li> </ul>               | End Users    |
| <b>Employee</b> | Responsible for managing and supporting different operations to ensure smooth food ordering, preparation, and delivery | <ul style="list-style-type: none"> <li>➤ Helps manage customer orders, update the menu, track inventory, assist with deliveries, and ensure food quality.</li> <li>➤ They also support customers by</li> </ul> | Staff        |



|  |           |                                                                   |  |
|--|-----------|-------------------------------------------------------------------|--|
|  | services. | answering queries and help generate basic reports for management. |  |
|--|-----------|-------------------------------------------------------------------|--|

### 3.3 User Environment

Using the SpiceSync system, users will feel comfortable managing and ordering fast food in a personalized, quick, and eco-friendly mode. Users like to have one easy access to real-time updates and informative options. SpiceSync will give users the possibility of designing a meal ingredient, tracking their orders, and partaking in loyalty programs without any hassle. The system is intent with varying nuances of users as far as tech knowledge goes, thus providing simple and intuitive interfaces along with encompassing multilingual support. Here is the idea of some suggestions:

- The main actors will be operators, Admins, Delivery Boys, Cooks, and Support Representatives in SpiceSync system for maintaining the operations and finishing tasks.
- As preferences change from time to time, the system shall keep flexibility. On regular bases update will be done, with an update cycle of about 3 weeks.
- Few unique environmental constraints maintain the cost of fresh, regional ingredients, handle increased order volume during peak hours, and ensure the delivery is faster than fast.
- This system will use java and also C++ to develop backend logic along with mobile application support.
- The typical fastfood ordering is perhaps nowadays a very busy thing. But in the near future, such things will evolve into a more futuristic sense of having holographic menu displays or AI voice bots for easier ordering and access.

The system can run on mobile devices, whether Android or iOS. It reads web browsers, like Chrome, Mozilla, and Safari. It will also read PCs and integrate fully with external delivery platforms like FoodPanda and Cheetay.

### 3.4 Stakeholder Profiles

Each stakeholder will be discussed in terms of their descriptions and assigned responsibilities in stakeholder profiles. In what manner is each stakeholder fulfilling their responsibility?

- **Managers** maintain the fastfood ordering system, update menus, run promotional deals (like Eid, Independence Day offers), and maintain service operations.

- **Admin** manages system configuration, monitors user accounts, promotional banners, and general security of the platform.
- **Users** interact with the system by registering or logging in, ordering based on meal preferences, real-time order tracking, participating in recipe contests, instant feedback, and earning loyalty reward points.
- **Chefs/ Kitchen staff** prepare customized meals in terms of spice levels and regional flavor choices and in doing so, meet standard quality and taste.
- **Delivery Riders** deliver the prepared food orders on time while promoting environmentalism.
- **System operators** ensure that the mobile application and the website are functioning both from the end-user point of view and the technical point of view on different devices, whilst integrating new user feedback and timely updates of the database system to uphold user satisfaction.

### 3.4.1 Manager

| Category                    | Particulars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Representation</b>       | Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b>          | Oversees daily operations within the restaurant, ensuring system efficiency and service quality.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Type</b>                 | Administrative Staff / Operations Lead                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Responsibilities</b>     | <ul style="list-style-type: none"> <li>➤ Menu updates and spice customization (i.e., Peshawari vs. Punjabi levels) within regions are managed.</li> <li>➤ Monitors AI-based meal recommendations and modifies them as per user feedback.</li> <li>➤ Coordinate with delivery partners (Foodpanda/Cheetah) for tracking real-time orders.</li> <li>➤ Resolve consumer complaints (e.g., wrong orders, payment issues).</li> <li>➤ Supervise helpdesk and chef staff to apply halal certification and transparency on ingredients.</li> </ul> |
| <b>Key Success Criteria</b> | <ul style="list-style-type: none"> <li>➤ Over 90% order accuracy through AI recommendations.</li> <li>➤ 20% increase in the number of subscription plans (office tiffins).</li> <li>➤ Positive ratings from users (&gt;4.5/5) on spice customization and delivery.</li> <li>➤ Complaint resolution time: less than 30 minutes.</li> </ul>                                                                                                                                                                                                   |
| <b>Involvement</b>          | <ul style="list-style-type: none"> <li>➤ Daily updates to the system (menu heading, promotions).</li> <li>➤ SpiceBot voice assistant performance review on a weekly basis.</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |

|                        |                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | ➤ Monthly review of festive promotion success (Eid/Ramadan deals).                                                                                                                     |
| <b>Deliverables</b>    | ➤ Monthly sales trend reports and user preference updates.<br>Updated halal certification records.<br>➤ Training modules for new staff on new features (for example, group discounts). |
| <b>Issues/Comments</b> | ➤ Expansion of multilingual support is essential (Sindhi/Balochi dialects).<br>➤ There is high demand for eco-friendly packaging, which requires us to negotiate with the vendor.      |

## 3.5 User Profiles

### 3.5.1 User

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>Representative</b>    | General user                                                        |
| <b>Description</b>       | Urban families, Office workers, Students ordering fast food online  |
| <b>Type</b>              | End-users                                                           |
| <b>Responsibilities</b>  | Register, Place orders, Rate meals, use SpicBot, Select spice level |
| <b>Success Criteria</b>  | frequent use, positive ratings, subscription purchases              |
| <b>Involvement</b>       | App/web use, order customization                                    |
| <b>Deliverables</b>      | Feedback, ratings, order data                                       |
| <b>Comments / Issues</b> | UI needs to be simple for non-tech savvy users                      |

## 3.6 Key Stakeholder or User Needs

| Need                                 | Priority | Concern                        | Current Solution | Proposed Solutions                         |
|--------------------------------------|----------|--------------------------------|------------------|--------------------------------------------|
| Personalized meal suggestions        | High     | Lack of custom flavor options  | Generic menus    | AI-based meal combo builder                |
| Halal ingredient transparency        | High     | No proof of ingredient quality | None             | Verified sourcing with halal certification |
| Fast and real-time delivery tracking | High     | Users can't see order status   | None             | Real-time GPS order tracking               |

|                            |        |                                |           |                                           |
|----------------------------|--------|--------------------------------|-----------|-------------------------------------------|
| Group orders and discounts | Medium | No shared ordering options     | None      | Group ordering module with bulk discounts |
| Voice-assisted ordering    | Medium | Users want easy access options | None      | SpiceBot integrated in Urdu and English   |
| Payment convenience        | High   | Limited payment options        | Cash only | JazzCash, Easypaisa, COD support          |

## 3.7 Alternatives and Competition

### 3.7.1 Cheezious

Cheezious is a fast food brand offering localized flavors but lacks real-time tracking, personalized suggestions or subscription plans.

### 3.7.2 McDonald's pakistan

International chain with a mobile app, but doesn't offer voice assisted ordering, regional spice adjustment, or cultural meal relevance.

## 4. Product Overview

The SpiceSync Fast Food Management System aims to streamline the online food ordering experience by introducing smart, AI powered, and culturally relevant features. The system offers dynamic interactions between end- users, restaurant partners, delivery agents, and administrative personnel. With mobile and web platform in sync, SpiceSync ensures that users receive real-time services, customizable meal options, eco-conscious delivery, and a loyalty reward system ----- all while promoting regional cuisine and culinary diversity.

### 4.1 Product Perspective

SpiceSync is a stand-alone system built with future integrations in mind, such as third-party delivery platforms and local vendor services. The system integrates multiple modules for order management, menu configuration, customer interaction, and reporting. It supports both android/iOS for mobile and is compatible with major web browser's on desktop.

**Table 4-1 Fast Food Management System**

| User Benefit                          | Supporting Features                                        |
|---------------------------------------|------------------------------------------------------------|
| Personalized fast food experience.    | AI-curated meal recommendations based on spice and region. |
| Efficient meal tracking               | Real-time GPS order tracking.                              |
| Convenience for working professionals | Subscription lunch/tiffin plans.                           |
| Transparent food sourcing             | Display of Halal certification and ingredients.            |
| Inclusive accessibility               | Urdu/English multilingual interface.                       |
| Easy payments                         | Integration with JazzCash, Easypaisa, and cash on delivery |
| Festive and cultural engagement       | Eid, Ramadan, Independence Day promotions                  |
| Quick feedback and issue resolution   | Instant order rating and help desk                         |
| Engagement and loyalty                | “Spice Points” reward program                              |
| Inclusive group ordering              | Bulk discounts and shared payment option                   |

## 4.2 Assumptions and Dependencies

- The System requires a stable Internet connection for real-time tracking and AI recommendations.
- Users must have basic mobile or browser access ( Android/IOS or Chrome/Google ).
- The system will rely on integration with platforms like Foodpanda or cheetay for third-party delivery.
- Ai recommendation logic assumes adequate user profile data ( preferences, locations, etc. ).
- Availability of regional ingredients may impact real time customization.

## 5. Product Features

### 5.1 Mobile Devices

The SpiceSync system will be accessible through Android and IOS platforms. Users can download the app from Google Play store or Apple app store and experience full access to all features including AI-curated meals, live tracking, and voice ordering.

## 5.2 Alerts

Alerts will be pushed through the app for:

- Order confirmation.
- Dispatch notification.
- Delivery updates.
- Subscription renewal.
- Promotion and discount alerts.
- Feedback requests.

## 6. Constraints

- Integration with multiple payment gateways and ensuring secure transactions.
- Ensuring accurate AI-based recommendations without using privacy violations.
- Adapting UI/UX to users with limited tech literacy.
- Insuring real-time data sync during peak hours and server load handling.
- Limited availability of eco-friendly packaging in certain regions.

## 7. Quality Ranges

### 7.1 Security:

End-to-end encrypted payment and personal data.

### 7.2 Availability:

99% uptime with redundancy support.

### 7.3 Usability:

Simple UI, support for Urdu and English.

### 7.4 Maintainability:

Modular code with update support every 3 weeks.

## 8. Precedence and Priority

The development will be split into two phases:

### 8.1 Phase 1:

Web platform development with core features ( registration, order tracking, payments, AI meal recommendations ).

## **8.2 Phase 2:**

Mobile app release with enhanced features like voice ordering, bulk ordering, and contest submissions.

# **9. Documentation Requirements**

## **9.1 User Manual**

The user manual will be available both as a printed guide and integrated within the app/web interface. It will cover account creation, meal customization, placing orders, and tracking deliveries.

## **9.2 Online Help**

A detailed online help section will be provided with FAQs, chat-bot support, and video tutorials.

## **9.3 Installation Guides, Configuration, and Read Me File**

A Read Me File and Setup Guide will be bundled with the web and mobile development packages, outlining installation instructions, required configurations, and version changes.

# **A Feature Attributes**

## **A.1 Status**

|              |                    |
|--------------|--------------------|
| Proposed     | Drone deliveries   |
| Approved     | SpiceBot           |
| Incorporated | AI recommendations |

## A.2 Benefit

|           |                    |
|-----------|--------------------|
| Critical  | Data Security      |
| Important | 24/7 uptime        |
| Useful    | Festive Promotions |

## A.3 Effort

Advanced features like AI recommendations and spice-bot requires high development effort, back-end integration, and testing. Simple modules like menus need less effort.

## A.4 Risk

Cost and timelines may vary due to third-party API integration and AI feature complexity. System load during peak hours is also a concern.

## A.5 Stability

The system is designed to be stable and modular. Only minor UI or content updates are expected after launch.

## A.6 Target Release

Beta version in July 2025; final version by October 2025 with full feature integration. Updates will follow a 3-week cycle.

## A.7 Assigned To

Development will be handled by back-end, front-end, AI team. The project manager will oversee the timeline and deliverables.

## A.8 Reason

These feature ensure personalization, cultural relevance, and system reliability ----- Key for user satisfaction and long-term use.



